# Tutorial 1 Notes cum Guidance

Hope you have all managed to access Oracle using Oracle client SQL Developer. (I suspect some are still missing the account information as there is still a mismatch between accounts issued and enrolled students). Please email Indu asap with your student ID and name after checking the Oracle account list from Stream). Everyone should change their password from the given default and make a note of it and keep it safe (to avoid it being getting locked or having to send a request to reset which may take a while).

VMware will allow you to access all Massey software. You need to use Massey username and password for this. The Oracle username and password is for accessing Oracle server via clients such as SQL Developer.

It is important you spend some time exploring the IDEs and it does take time before you get comfortable in using. We strongly recommend exploring both JDeveloper and Visio for modelling, later you can pick one you prefer. The Tutorial 1B shows just one way to explore, feel free to experience and explore the way you prefer.

Tutorial 1A – part B – make sure the file path doesn't have spaces or wild characters. Also need to extract the file and provide the correct/complete path (sql file is under a folder(s)).

Tutorial 1A – part C had 8 questions where the data for two tables is closely analysed to learn about some of the concepts as below:

- table/file, record/row, field/column.
- looking at file structure, data (itself)
- changing the file structure for query requirements, data redundancies, anomalies, etc.
- mapping of data with core concepts (from domain); - gives you an idea about better design.
- using/value of good naming conventions (help in understanding, maintenance, etc.)

Some further comments about some of the questions:

Q2. The city names are contained within the MANAGER_ADDRESS attribute (look at the data closely and pick a city or two). Think of the problems that may occur if you were to generate a city wise list (say for 100000s of records).

- Would it be easy to extract city from given long string of MANAGER_ADDRESS?
- Ease in writing queries (coding), time taken to run such queries (which are doing extraction as well).

Q4. Even in such a small table, you will notice so much unnecessary repetition. Note that the manager named Holly B. Parker occurs three times, what else is repeating because of this? What if she changes her address and phone number?

What if she is a long-term employee and changes her address 20 times. Remember she can be working for say an average of 2-3 projects at the same time. Given these situations so much data would unnecessarily need to be stored. What happens over a period and each time a change (update) is made? If any occurrence is missed during the change, the data are "different" for the same person. After some time, it may become difficult to determine what the correct data are. In addition, multiple occurrences invite misspellings and digit transpositions, thus producing some further anomalies.

There can be multiple employees like her compounding the redundancy problem.

Q5. Look at the data closely for redundancies. Did you notice hourly rate for the CT job code (60, 60, 62 and 26). What happened there? What are the repercussions of such mistakes (we know the reason, bad design)?

Q7. Think of possible data sources. The naming conventions will give you some clue but that may also be occasionally misleading. Proj_Hours – what do they represent? Do they belong to the project as indicated by the prefix or do they belong to the employee (which is missing from the col-name)? They are the hours worked by an employee for a project, so probably map to (dual) concept Emp_Proj (there may be a domain name for such a table/concept e.g., Charge or we can leave Emp_Proj as well). Hope you have spotted all 4 concepts/data sources.